

Problem 33. Three circular arcs w_1, w_2, w_3 with common endpoints A and B are on the same side of the line AB ; w_2 lies between w_1 and w_3 . Two rays emanating from B intersect these arcs at M_1, M_2, M_3 and K_1, K_2, K_3 , respectively. Prove that

$$\frac{M_1 M_2}{M_2 M_3} = \frac{K_1 K_2}{K_2 K_3}.$$

Solution. From inscribed angles we have $\angle AK_1 B = \angle AM_1 B$ and $\angle AK_2 B = \angle AM_2 B$. From this it follows that $\triangle AK_1 K_2 \sim \triangle AM_1 M_2$, so

$$\frac{K_1 K_2}{M_1 M_2} = \frac{AK_2}{AM_2}.$$

Similarly $\triangle AK_2 K_3 \sim \triangle AM_2 M_3$, so

$$\frac{K_2 K_3}{M_2 M_3} = \frac{AK_3}{AM_3}.$$

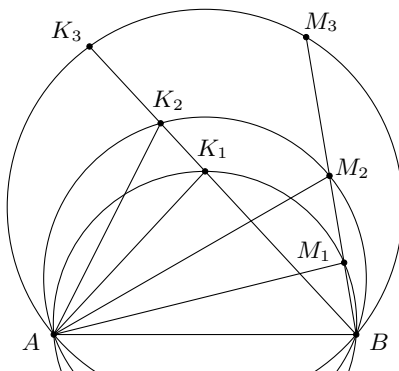
From these equations we get

$$\frac{K_1 K_2}{M_1 M_2} = \frac{K_2 K_3}{M_2 M_3},$$

from which the desired property follows.

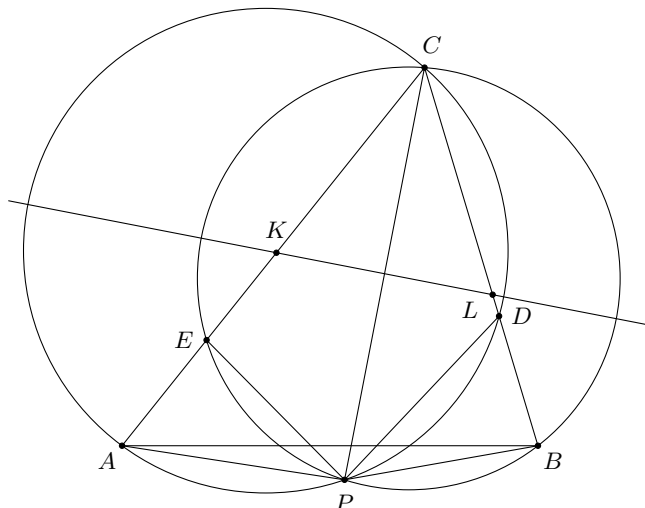
Alternatively, it is well-known that there exists a spiral similarity with centre A mapping M_1 to K_1 and M_2 to K_2 . The same spiral similarity also maps M_3 to K_3 by considering the circles w_2 and w_3 . So the figures $M_1 M_2 M_3$ and $K_1 K_2 K_3$ are similar. This implies

$$\frac{M_1 M_2}{M_2 M_3} = \frac{K_1 K_2}{K_2 K_3}.$$



Problem 34. Let the points D and E lie on the sides BC and AC , respectively, of $\triangle ABC$, satisfying $BD = AE$. The line joining the circumcentres of $\triangle ADC$ and $\triangle BEC$ meets the lines AC and BC at K and L , respectively. Prove that $KC = LC$.

Solution.



Assume that the circumcircles of triangles ADC and BEC meet at C and P . The problem is to show that the line KL makes equal angles with the lines AC and BC . Since the line joining the circumcentres of triangles ADC and BEC is perpendicular to the line CP , it suffices to show that CP is the angle bisector of $\angle ACB$.

Since the points A, P, D, C are concyclic, we obtain $\angle EAP = \angle BDP$. Analogously, we have $\angle AEP = \angle DBP$. These two equalities together with $AE = BD$ imply that triangles APE and DPB are congruent. This means that the distance from P to AC is equal to the distance from P to BC , and thus CP is the angle bisector of $\angle ACB$, as desired.

Note that one can easily deduce $\triangle APE \sim \triangle DPB$ from the spiral similarity configuration.

□

Problem 35. Let m and n be relatively prime positive integers. Determine all possible values of

$$\gcd(2^m - 2^n, 2^{m^2+mn+n^2} - 1).$$

Answer. 1 and 7

Solution. Without loss of generality we may assume that $m \geq n$. It is well known that

$$\gcd(2^p - 1, 2^q - 1) = 2^{\gcd(p,q)} - 1$$

so

$$\gcd(2^m - 2^n, 2^{m^2+mn+n^2} - 1) = \gcd(2^{m-n} - 1, 2^{m^2+mn+n^2} - 1) = 2^{\gcd(m-n, m^2+mn+n^2)} - 1.$$

Let $d \geq 1$ be a divisor of $m - n$. Clearly, $\gcd(m, d) = 1$, since m and n are relatively prime. Assume that d also divides $m^2 + mn + n^2$. Then d divides $3m^2$. In view of $\gcd(m, d) = 1$ the only choices for d are $d = 1$ and $d = 3$. Hence $\gcd(m - n, m^2 + mn + n^2)$ is either 1 or 3. Consequently, $\gcd(2^m - 2^n, 2^{m^2+mn+n^2} - 1)$ may only assume the values $2^1 - 1 = 1$ and $2^3 - 1 = 7$. Both values are attainable, since $m = 2, n = 1$ gives

$$\gcd(2^2 - 2^1, 2^{2^2+2 \cdot 1 + 1^2} - 1) = \gcd(2, 2^7 - 1) = 1$$

whereas $m = 1, n = 1$ gives

$$\gcd(2^1 - 2^1, 2^{1^2+1 \cdot 1 + 1^2} - 1) = \gcd(0, 2^3 - 1) = 7.$$

□

Problem 36. A directed graph does not contain directed cycles. The number of edges in any directed path does not exceed 99. Prove that it is possible to colour the edges of the graph in 2 colours so that the number of edges in any single-coloured directed path in the graph will not exceed 9.

Solution. Label each vertex by the number from 0 to 99, that is equal to the length of the longest directed path that ends in this vertex (so two vertices may have the same label). Then every edge goes from a vertex with a smaller label to a vertex with a larger label. Now colour this edge in red if the digit of tens in the larger label is greater than the digit of tens in the smaller label. Otherwise colour this edge in blue. Since the number of tens is the same in all vertices on a blue path, the length of the path cannot exceed 9. Since the number of tens is different in all vertices on a red path, the length of the path also cannot exceed 9.

□